

Road network characteristics and Land Surface Temperature in two South Indian cities: Bangalore and Chennai.

Ellen Brock

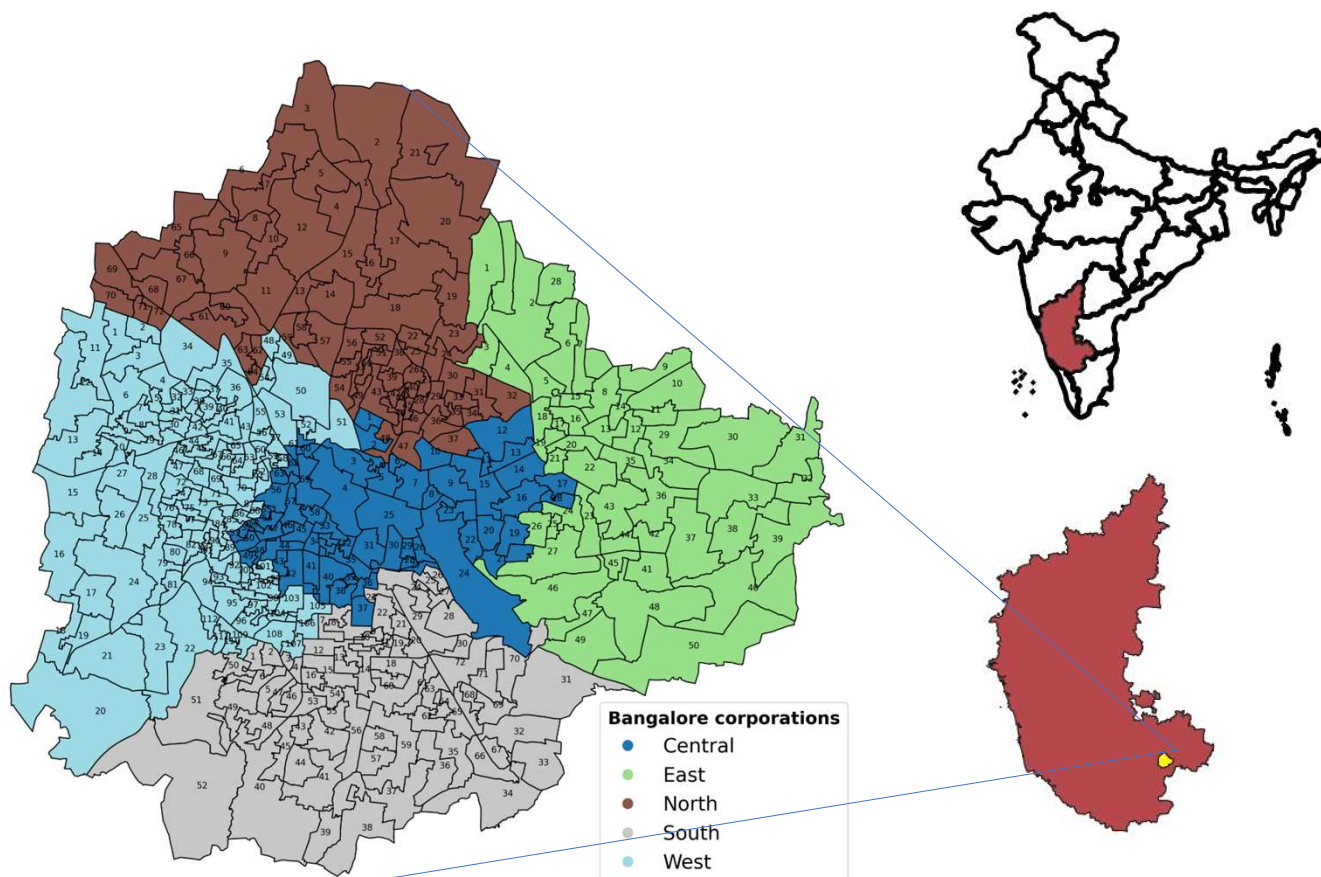
Agenda

1. Motivation
2. Ward level data
3. Computation of Land Surface Temperature (LST)
4. Computation of road characteristics
5. Computation of other characteristics
6. Correlation and regression analysis
7. Code
8. Conclusion + next steps

Motivation

- Very **few studies** that have looked at road characteristics in general and Land Surface Temperature (LST) and road characteristics in particular
 - Exception: Mathew et al. (2020) for Ahmedabad who look day and night Land Surface Temperature (LST) using MODIS data with digitized road data
- This work here uses **Landsat data** and **automatically extracts roads** and certain road characteristics using Python next to road density
- **Mixed results in the literature** in general for the link between certain road characteristics/metrics and LST
- **Bangalore and Chennai**: ward level analysis

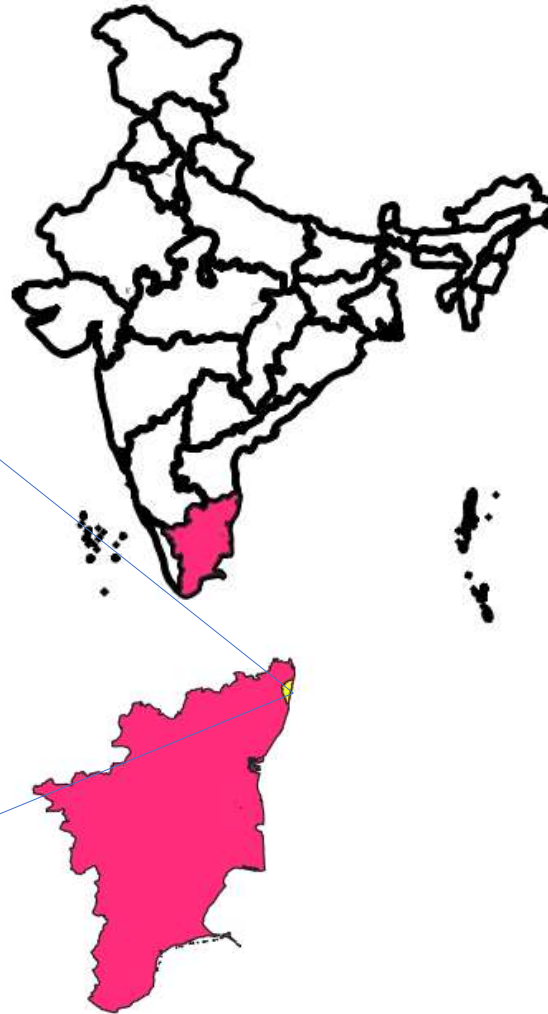
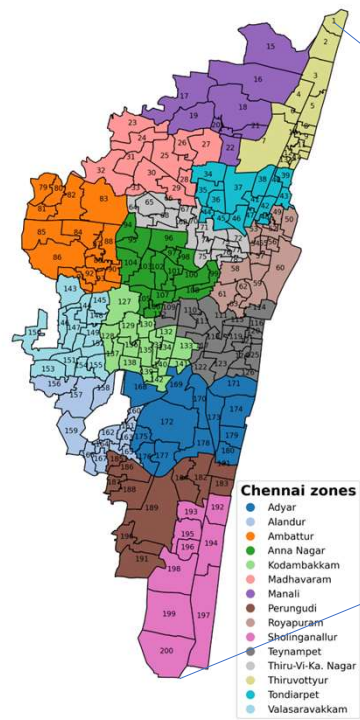
Bangalore



Source: Own computations based on <https://github.com/datameet/maps> and Opencity

- **Ward Level Climate Action Plans (WCAPs)** to build local roadmaps for climate action
- Climate Action Cell will develop ward action plans for ten wards in five city corporations of Bengaluru and then roll-out to other wards
- **5 zones/corporations**
- **369 wards**

Chennai



- Chennai is divided into **15 zones**
- A total of **200 wards**

Source: <https://chennai.nic.in/about-district/administrative-setup/local-bodies/>

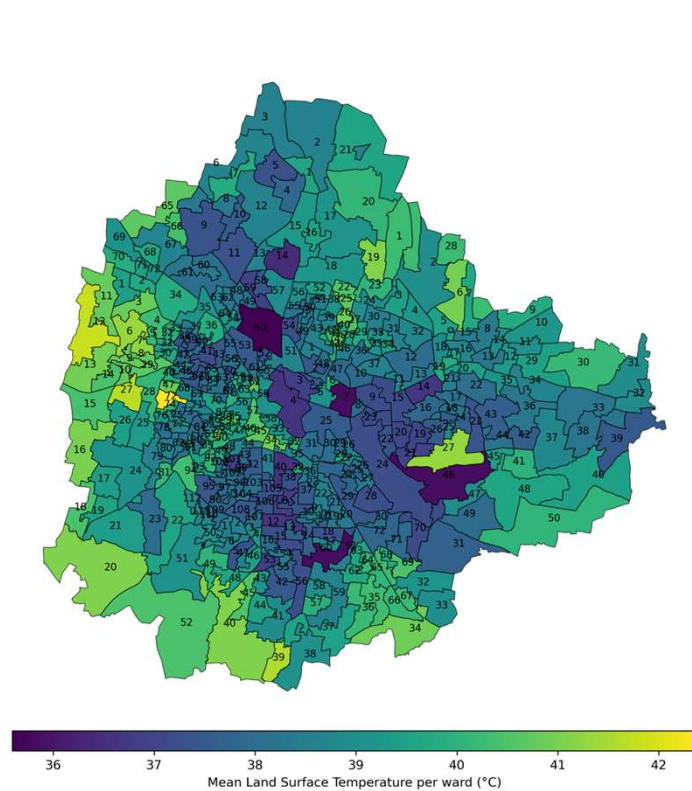
Data clean up – ward level data

- Python has some functions to clean up the data for invalid polygons but it was not sufficient
- **QGIS** and **map shaper** were used based on the YouTube tutorials of Ujaval Gandhi

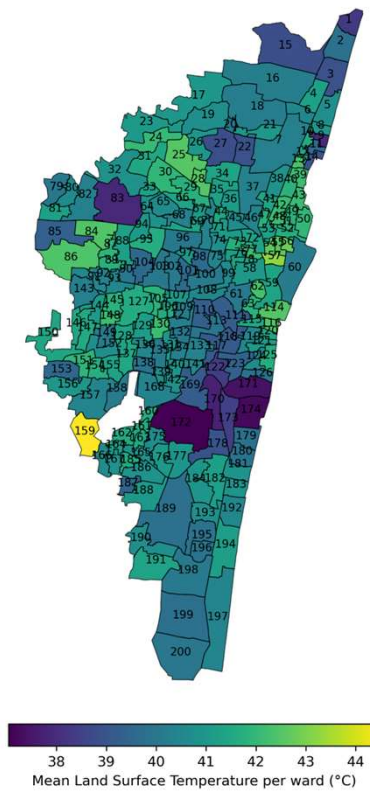
Land surface temperature

- Fully computed in Python and for the summer months (March – May) for the years 2024-2025 using a **median composite** using **Landsat 8**
- LST computed in **three different ways**:
 - Band 10 thermal band only with scaling factor and cloud masking
 - Land Surface Temperature calculated from scratch from Landsat 8
 - Ermida et al. (2020) in JavaScript Python
- The first two methods were in geemap
- Conversion into degrees Celsius
- **Cloud masking** is used
- Converted to Celsius
- The results are robust regarding the three methods.
- **Band 10 thermal** band is used in the next slides

Land surface temperature (band 10) – contd.



Bangalore



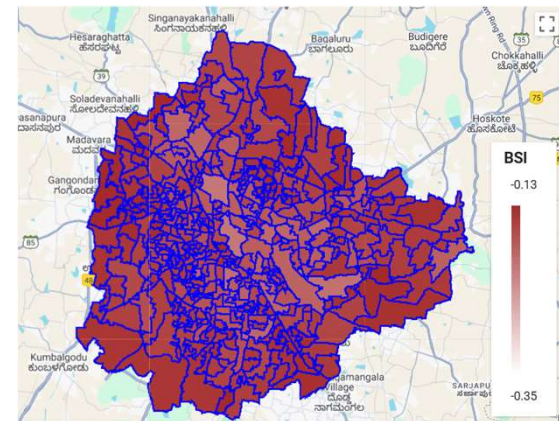
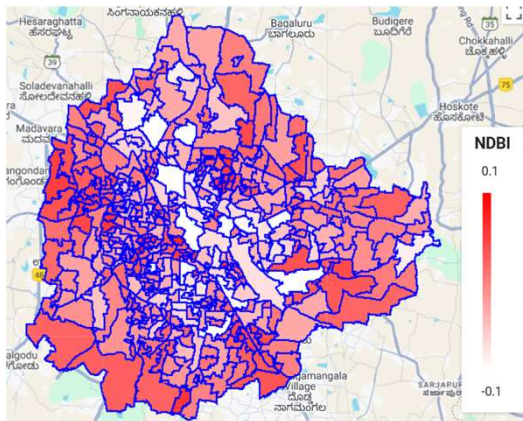
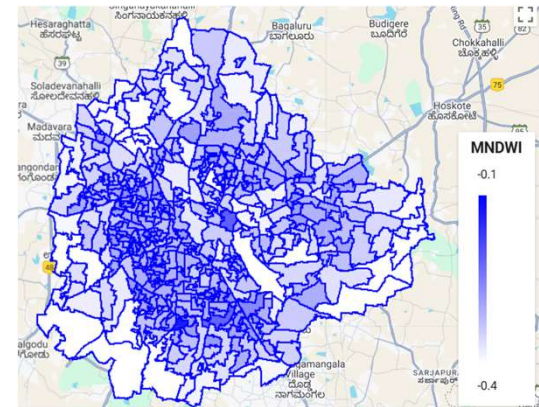
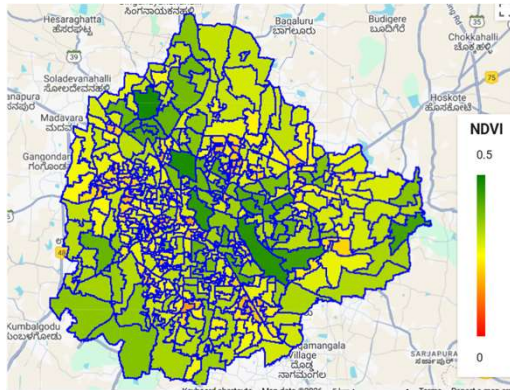
Chennai

Indices

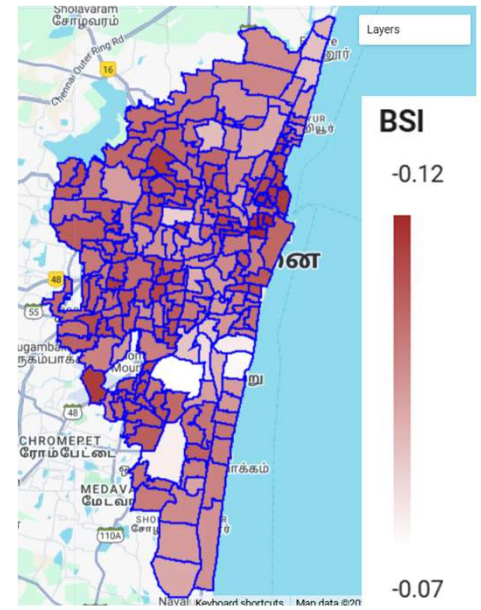
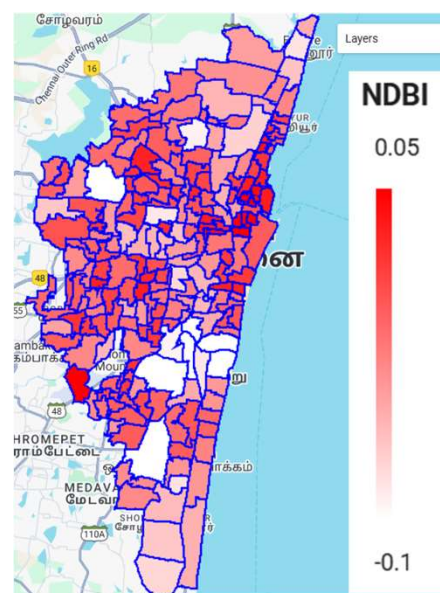
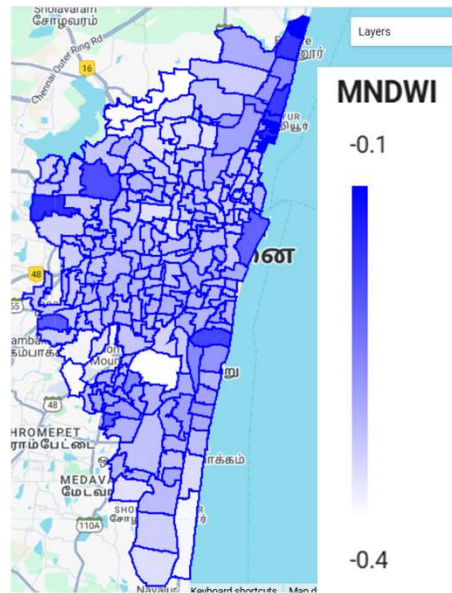
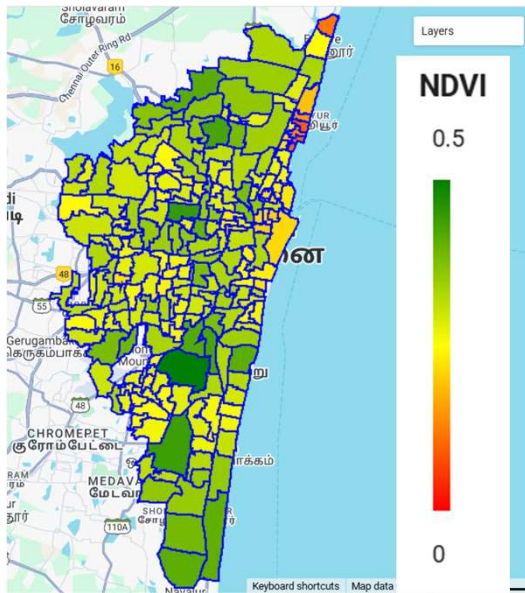
Name	Formula	Formula (bands (B) Landsat 8)
Normalised Difference Vegetation Index (NDVI)	$\frac{NIR - RED}{NIR + RED}$	$\frac{B5 - B4}{B5 + B4}$
Modified Normalised Difference Water Index (MNDWI)	$\frac{GREEN - SWIR1}{GREEN + SWIR1}$	$\frac{B3 - B6}{B3 + B6}$
Normalised Difference Building Index (NDBI)	$\frac{SWIR1 - NIR}{SWIR1 + NIR}$	$\frac{B6 - B5}{B6 + B5}$
Bareness Soil Index (BSI)	$\frac{(SWIR1 + RED) - (SWIR1 + RED)}{(SWIR1 + RED) + (SWIR1 + RED)}$	$\frac{(B6 + B4) - (B6 + B4)}{(B6 + B4) + (B6 + B4)}$

- Computed in Python using geemap using Landsat 8
- **Zonal statistics** at the ward level

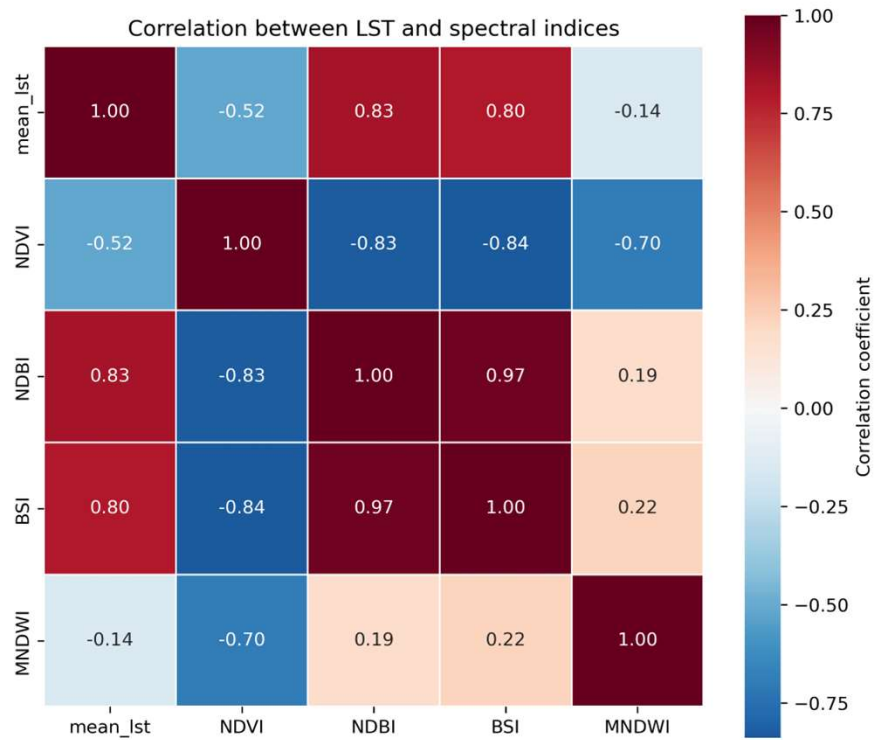
Bangalore: Indices



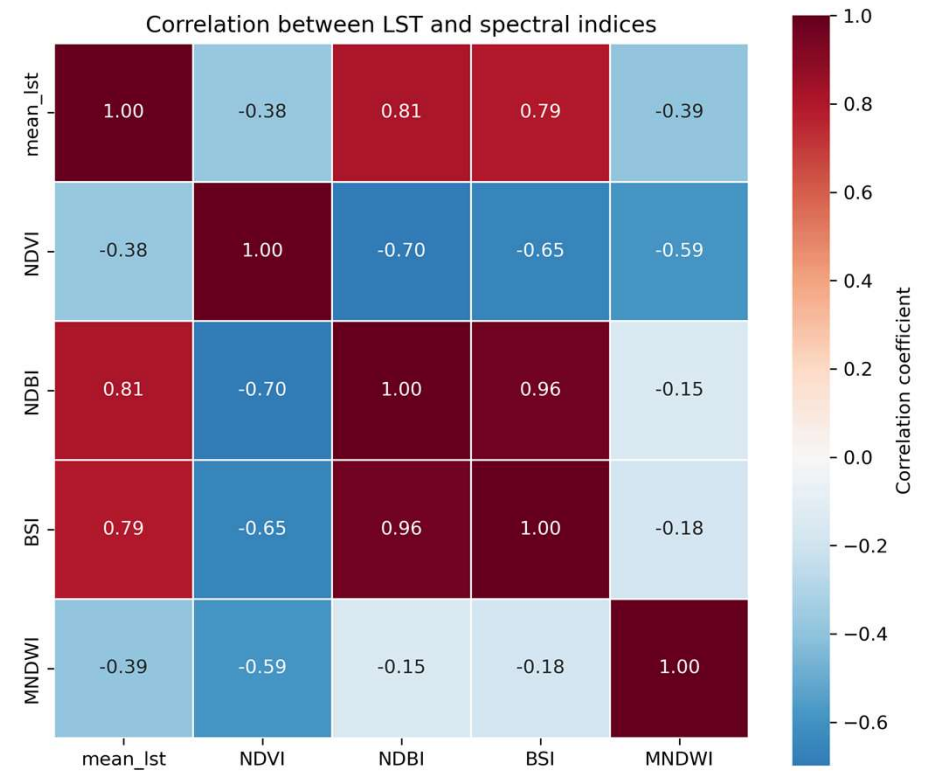
Chennai indices



Indices - correlation

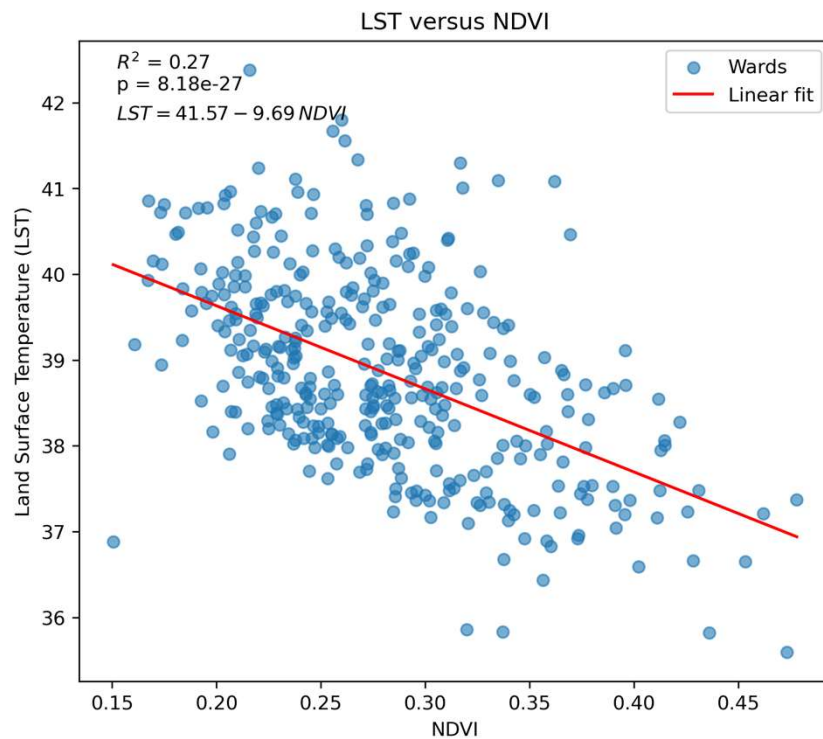


Bangalore

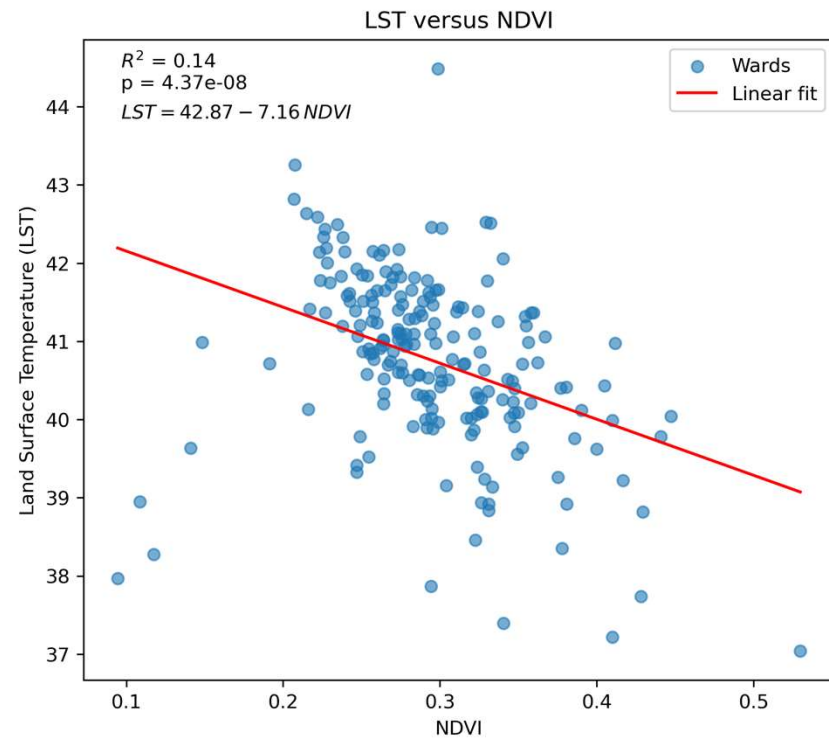


Chennai

Normalised Difference Vegetation Index (NDVI): regression

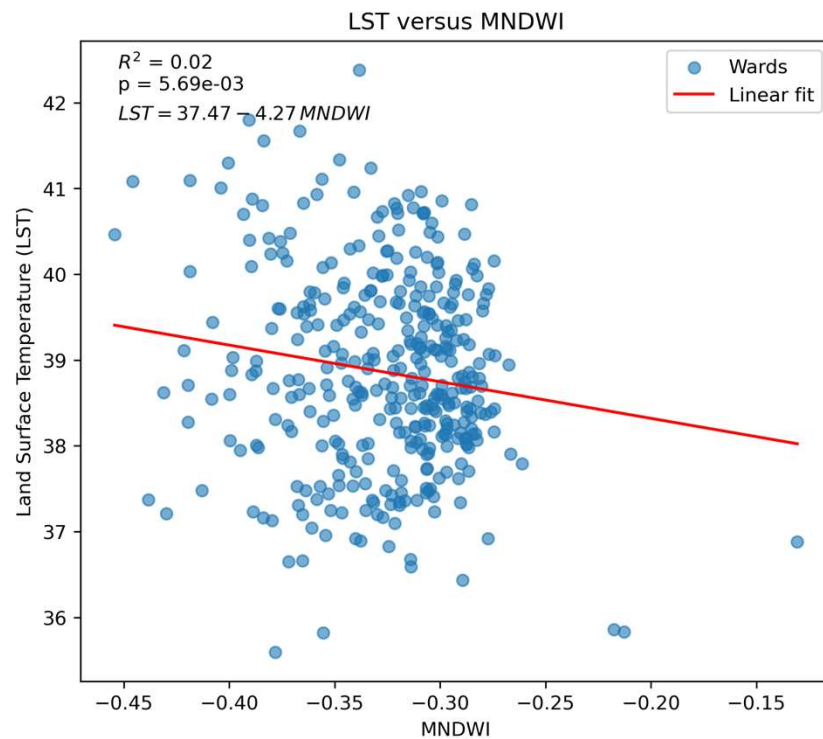


Bangalore

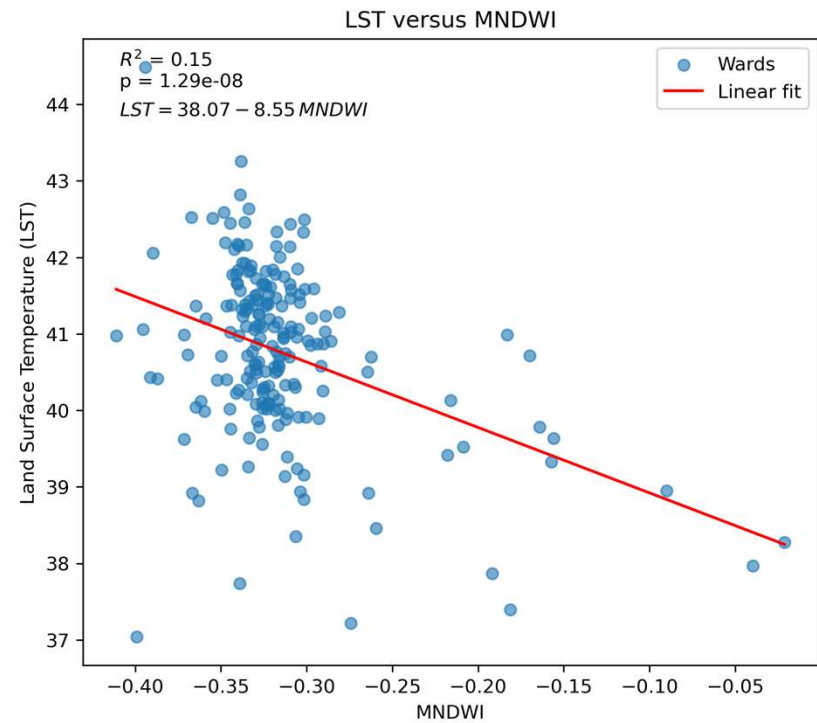


Chennai

Modified Normalised Difference Water Index (MNDWI): regression

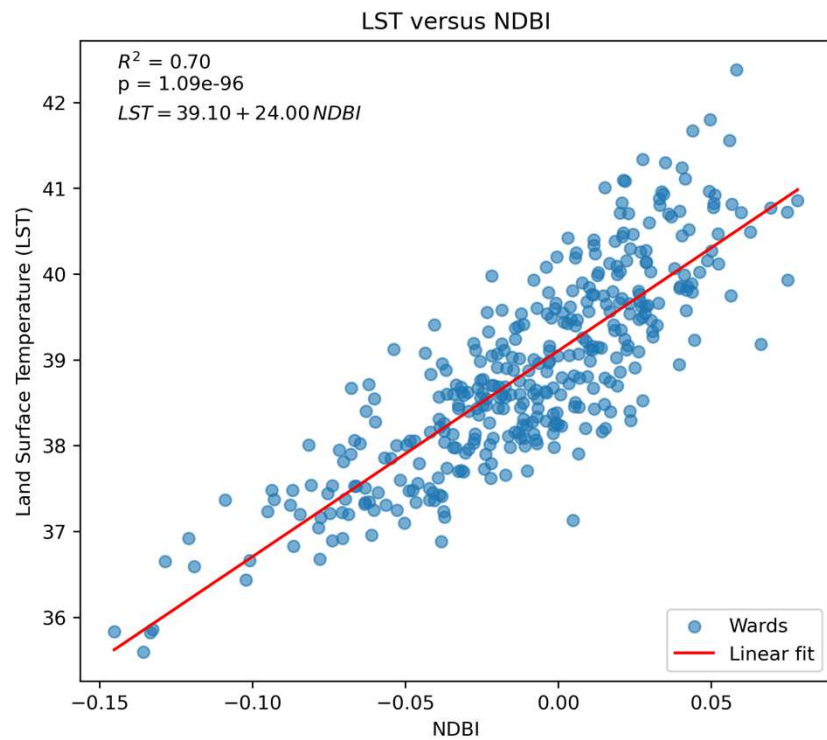


Bangalore

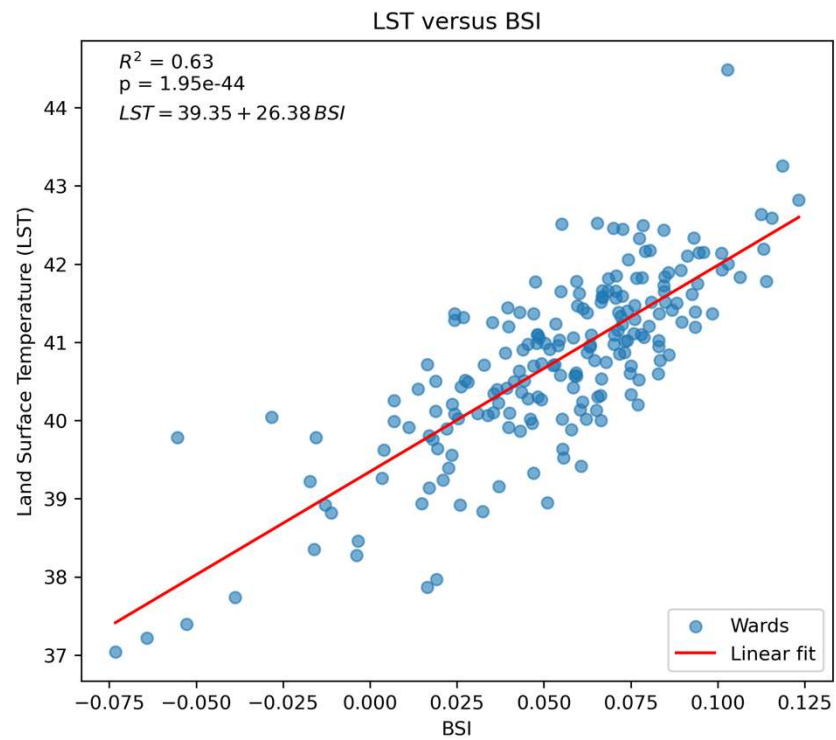


Chennai

Normalised Difference Building Index (NDBI): regression

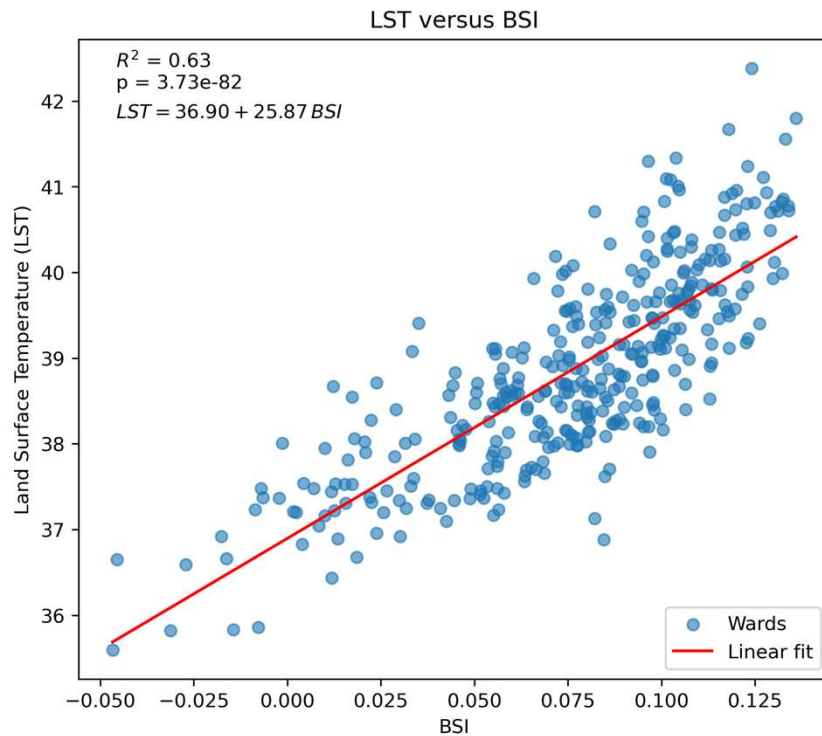


Bangalore

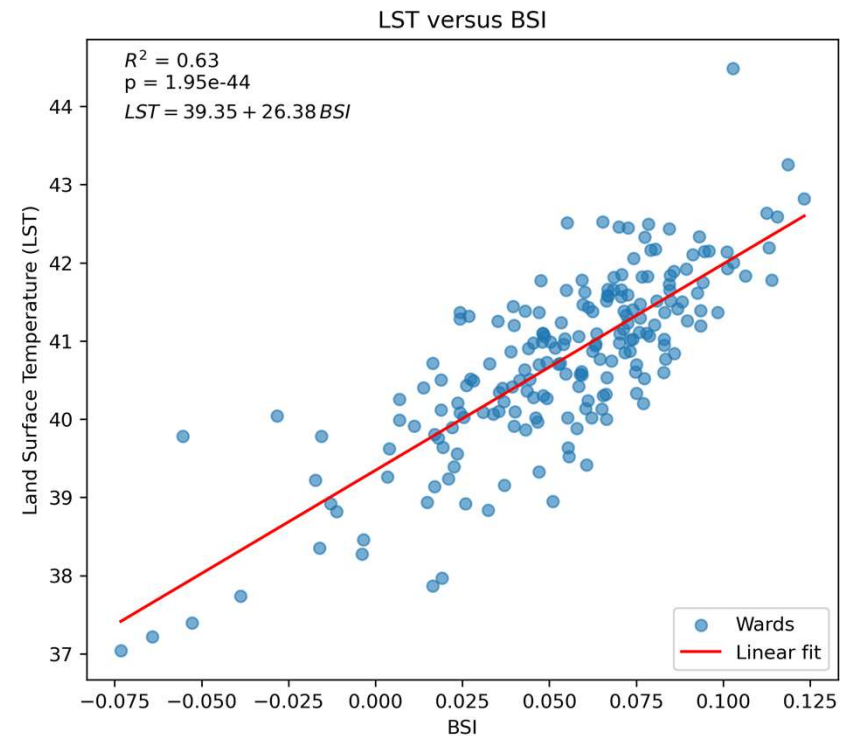


Chennai

Bareness Soil Index (BSI): regressssion



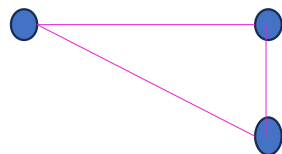
Bangalore



Chennai

Road indices

Metric	Formula	Description
Road density	$\frac{\text{street length (metres)}}{\text{area (km}^2\text{)}}$	Computed at the ward level using the area of the ward
Straightness (nodes)	$\frac{1}{n-1} \sum_{j \in N, j \neq i} \frac{d_{ij}^{Eu}}{d_{ij}}$	d_{ij}^{Eu} : Euclidian distance between nodes i and j d_{ij} : distance between nodes i and j n :
Global closeness	$\frac{n-1}{\sum_{v=1}^{n-1} d(v, u)}$	Average shortest path distance to u over all $n-1$ nodes within reachable nodes.
Local closeness	Closeness but with a radius of 1000 metres	



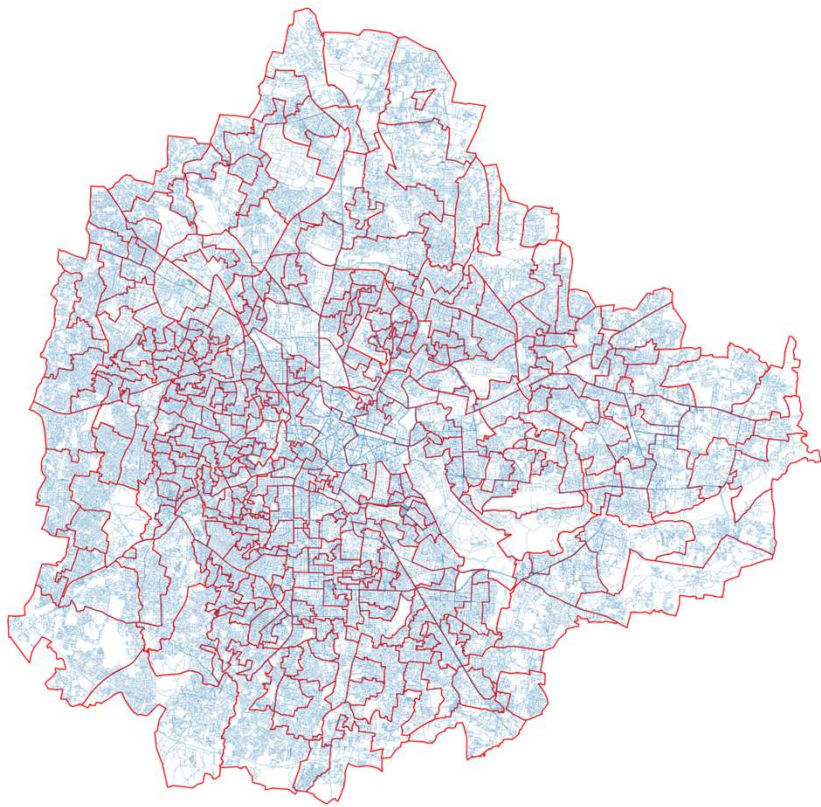
Nodes



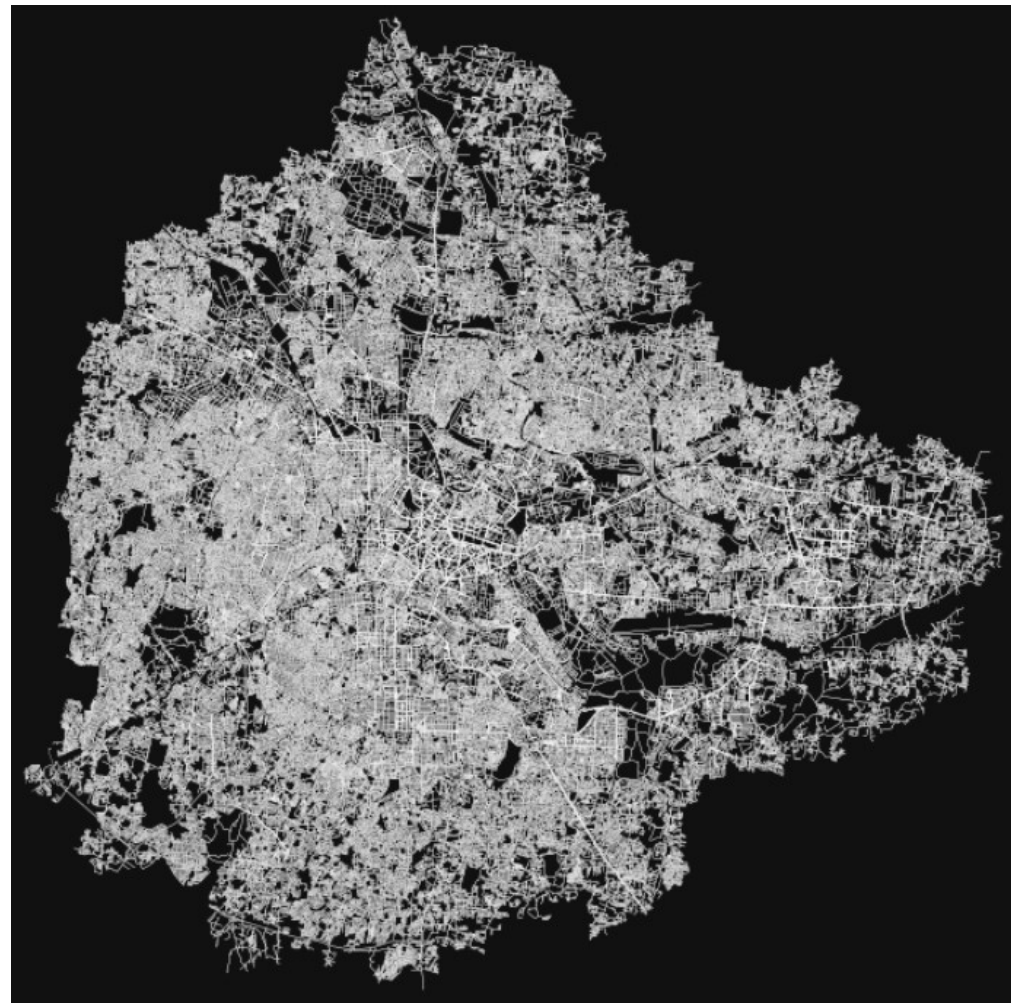
Edges (streets)

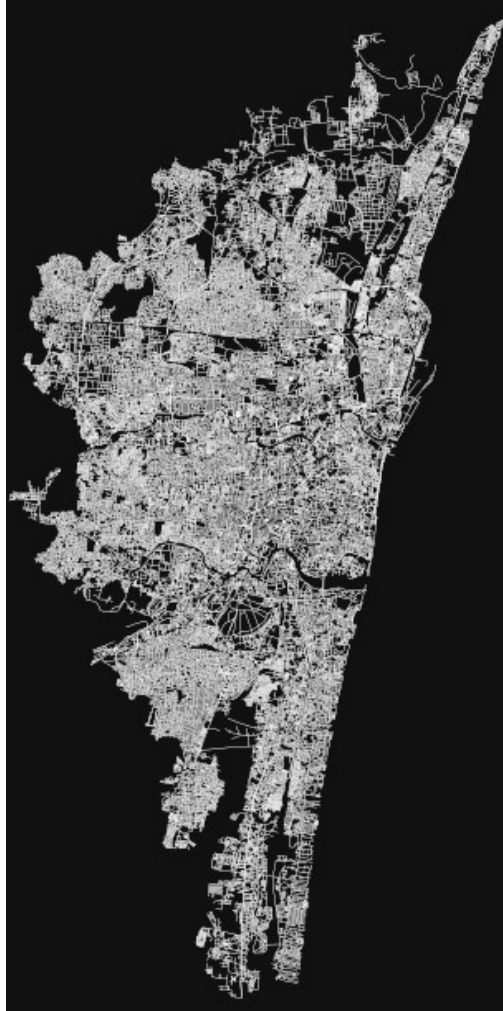
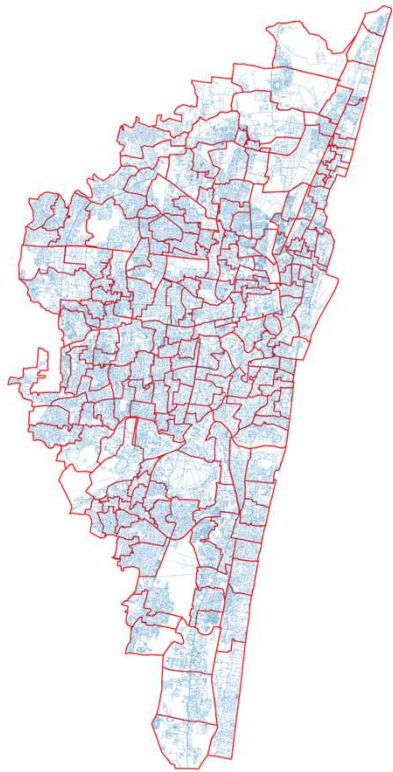
Road indicators

- Used the **osmnx** Python package to:
 - Extract the road network
 - To compute the street density
- Street network per ward extracted through “**truncate**” in **osmnx** downloading the entire city network (rather than making 369 “calls” to Overpass)
- All roads versus **drivable roads** only
 - One can also use drivable roads and service roads (in progress)
- Use of the Python **momepy package** to compute straightness, closeness and betweenness centrality



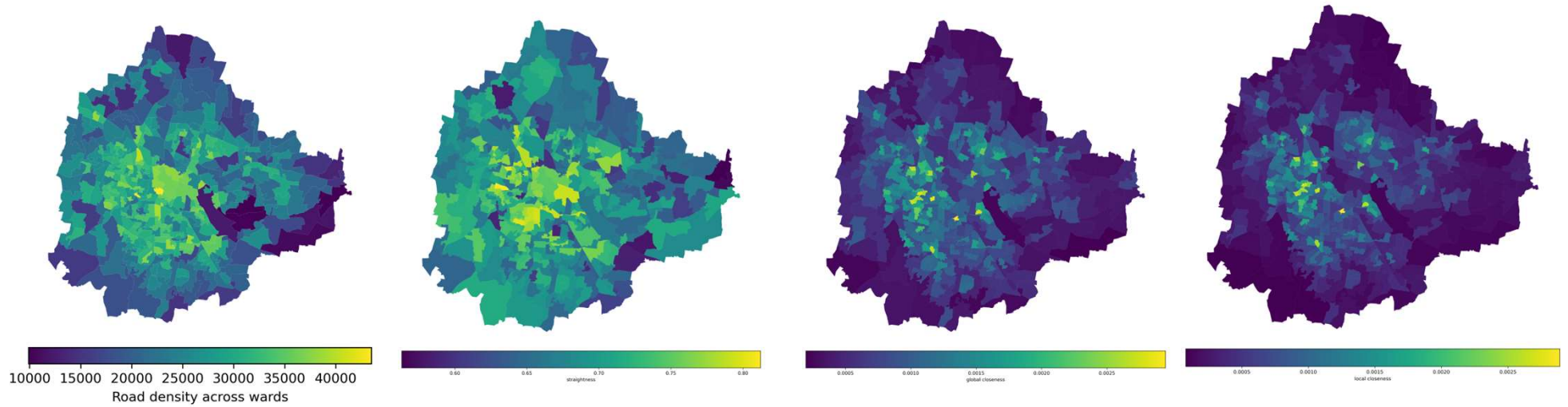
Bangalore has an area of **717.21 km²**
The total street length is **17 638.74 km**.
Average density is **24.59 km** per square kilometre





The total street length is 8417.18 km
Chennai has a total area of 431.96 km²
The street density is 19.49 km per km²

Road indicators Bangalore



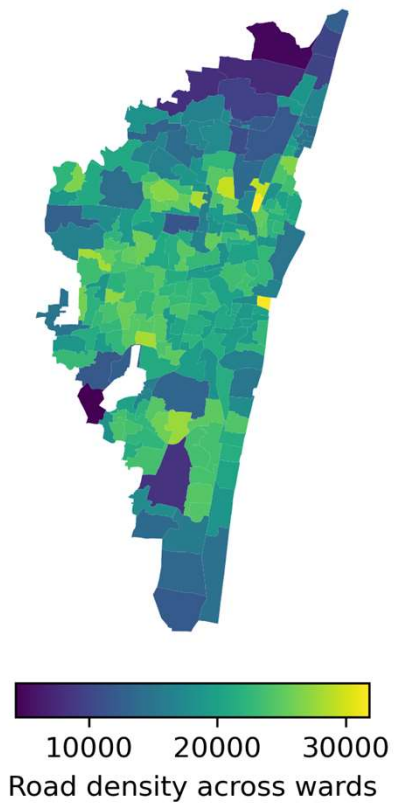
Road density

Straightness

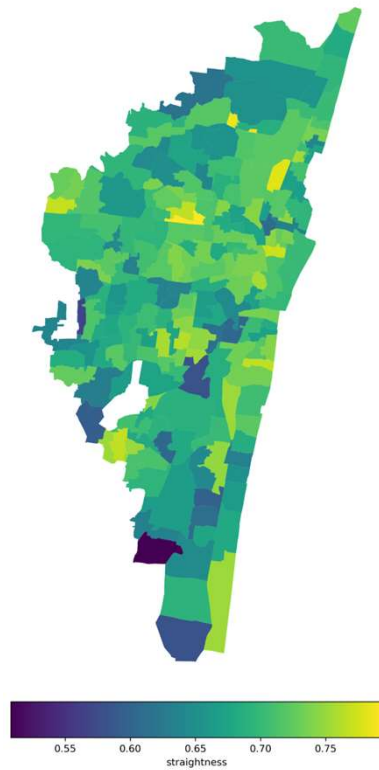
Global closeness

Local closeness

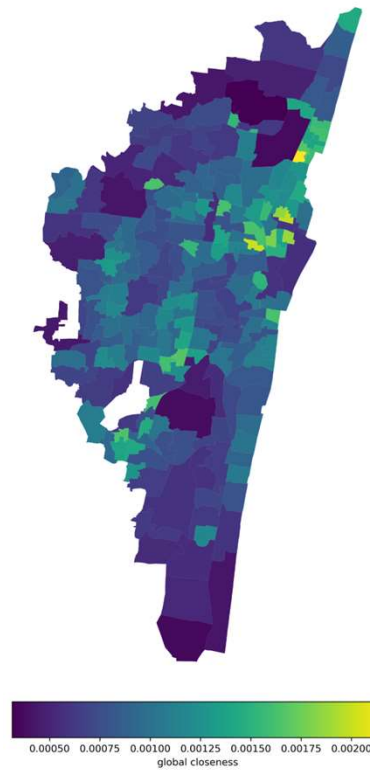
Road indicators: Chennai



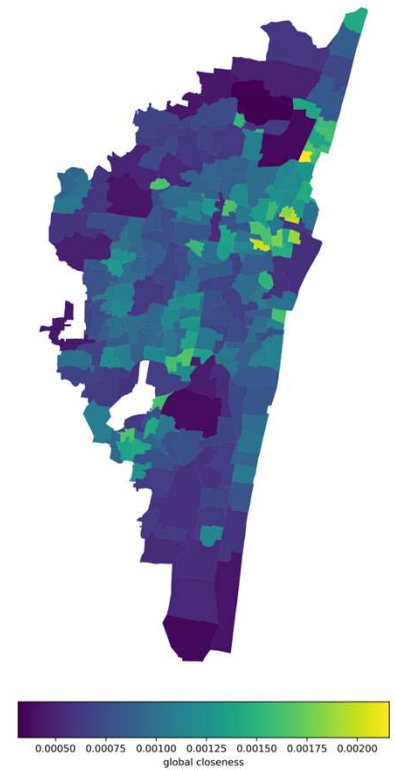
Road density



Straightness

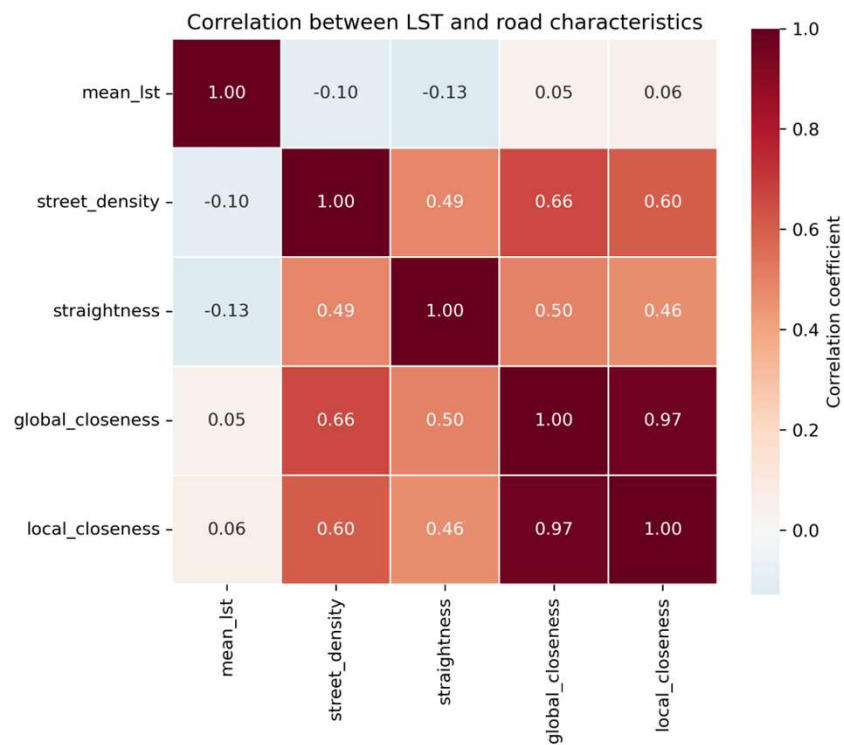


Global closeness

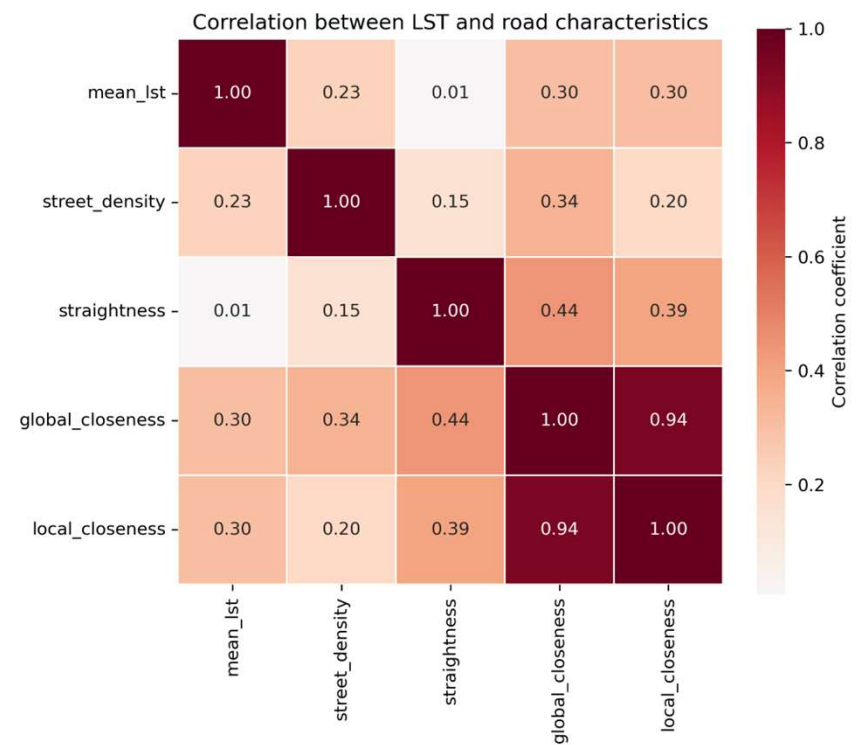


Local closeness

Correlation (all roads)

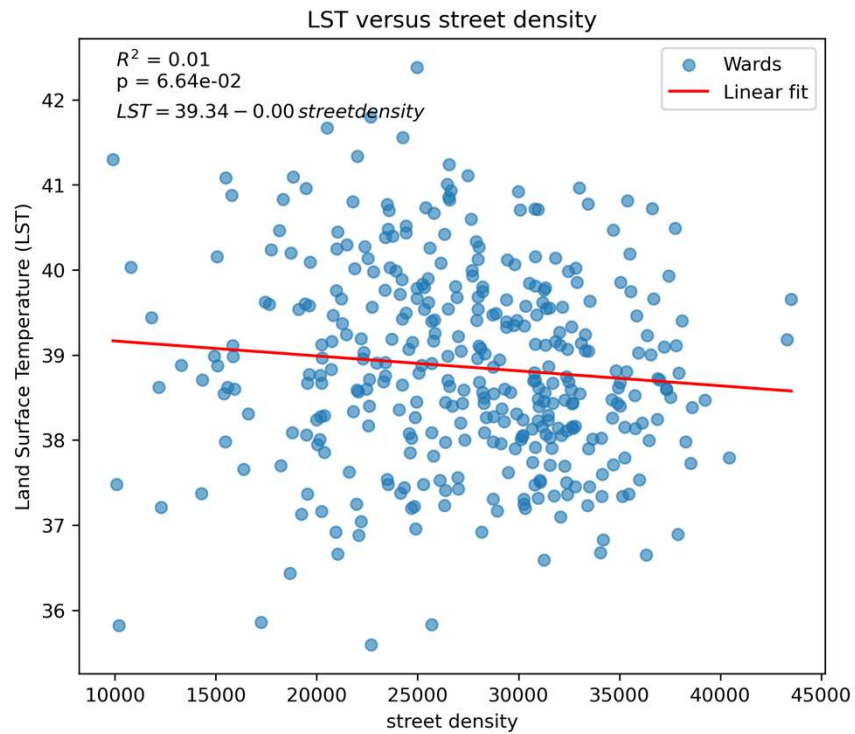


Bangalore

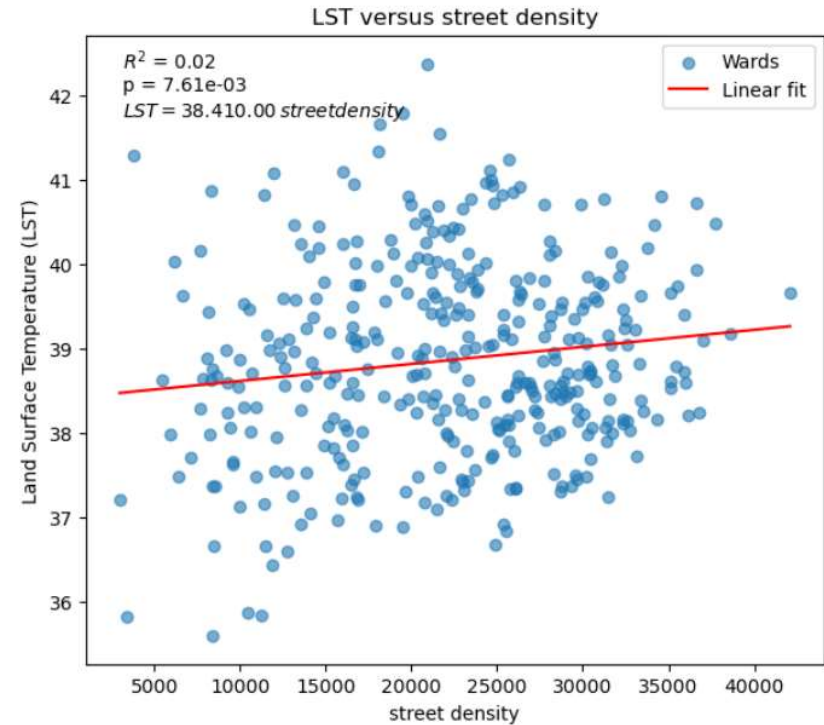


Chennai

Road density (Bangalore): regression

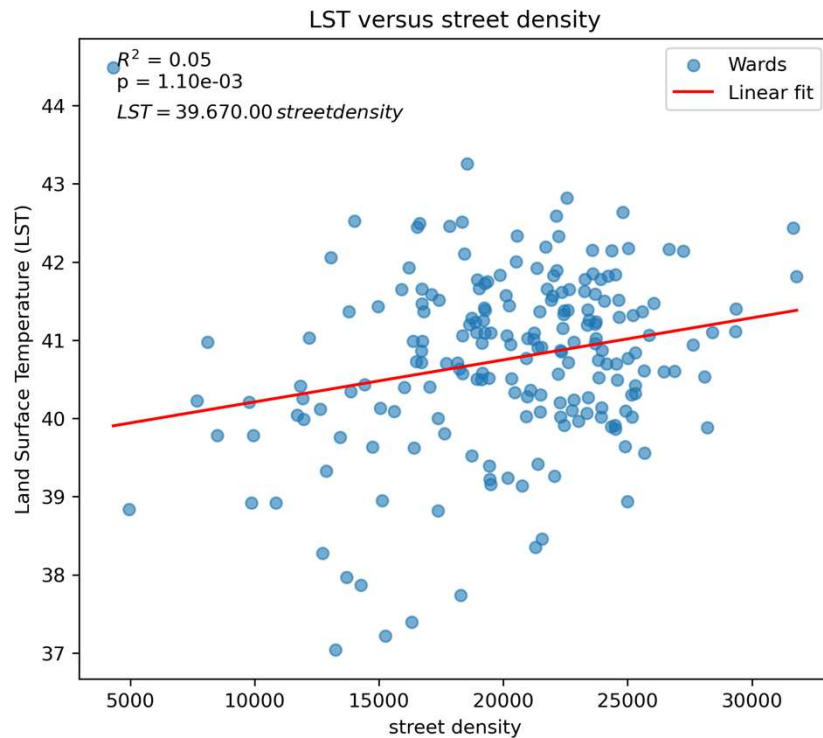


All roads

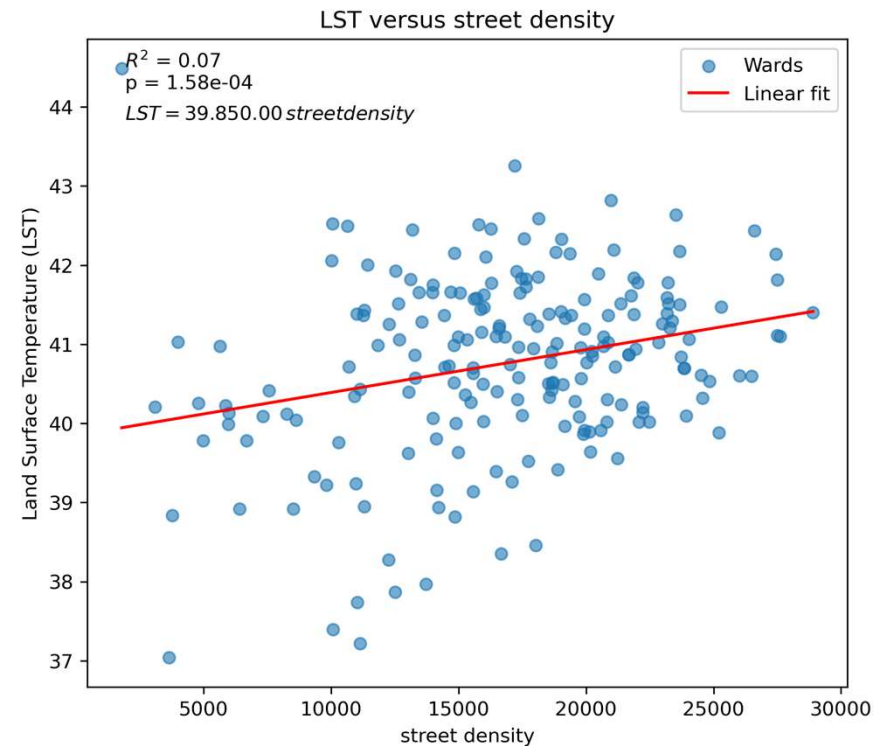


All drivable roads

Road density (Chennai): regression

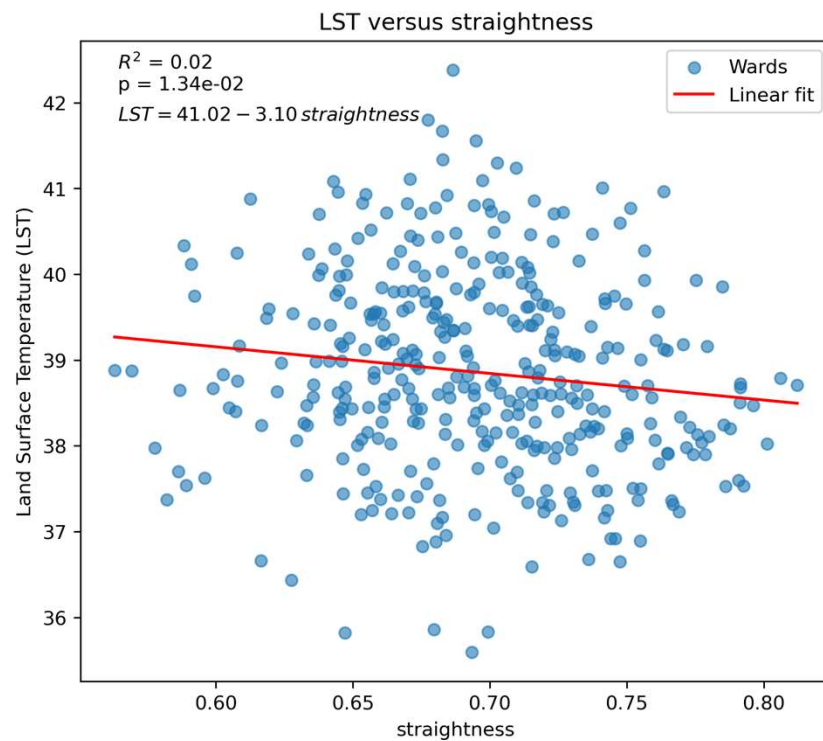


All roads

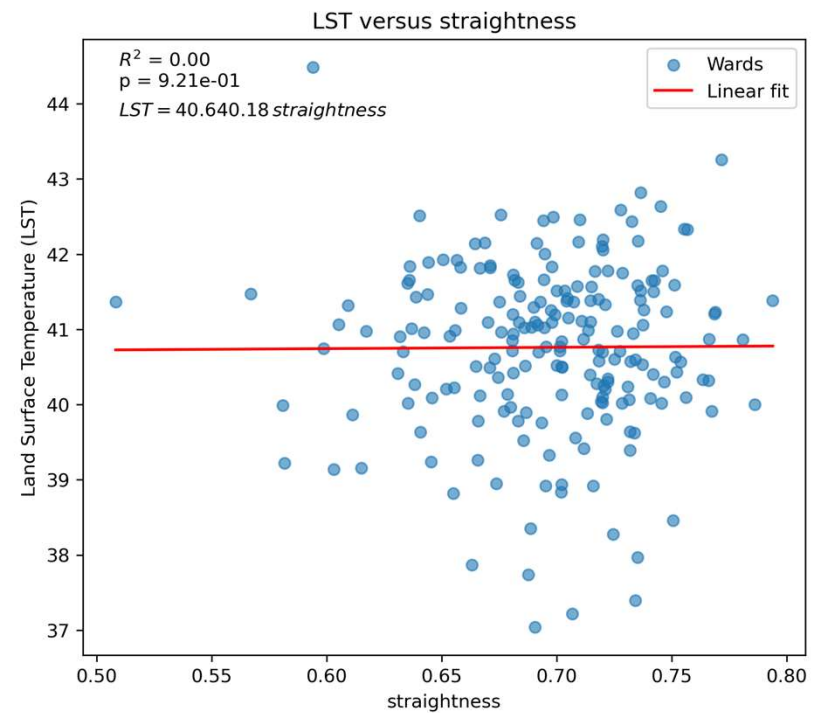


All drivable roads

Straightness: regression (all roads)

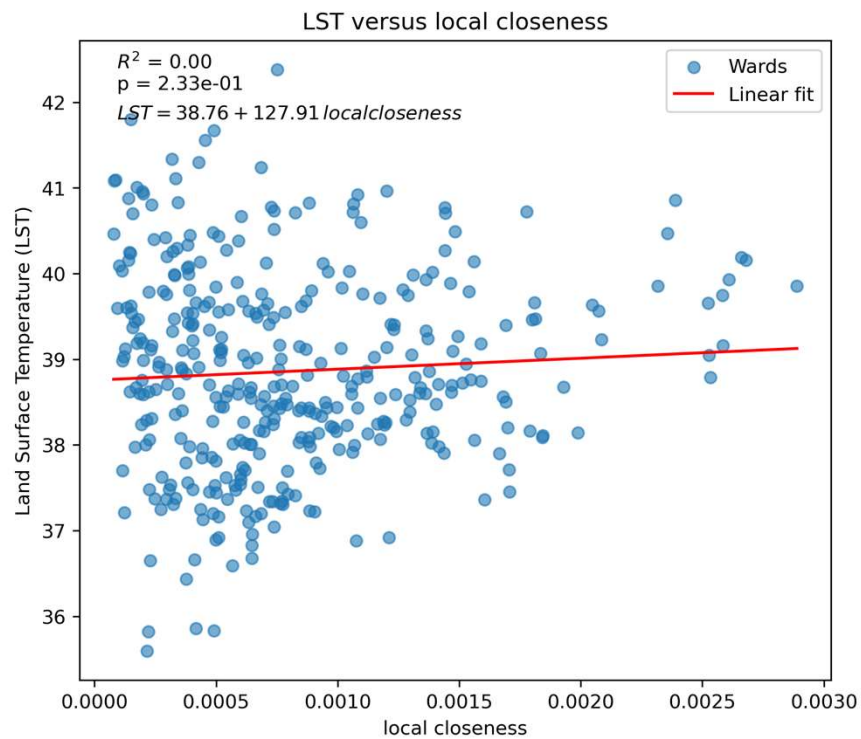


Bangalore

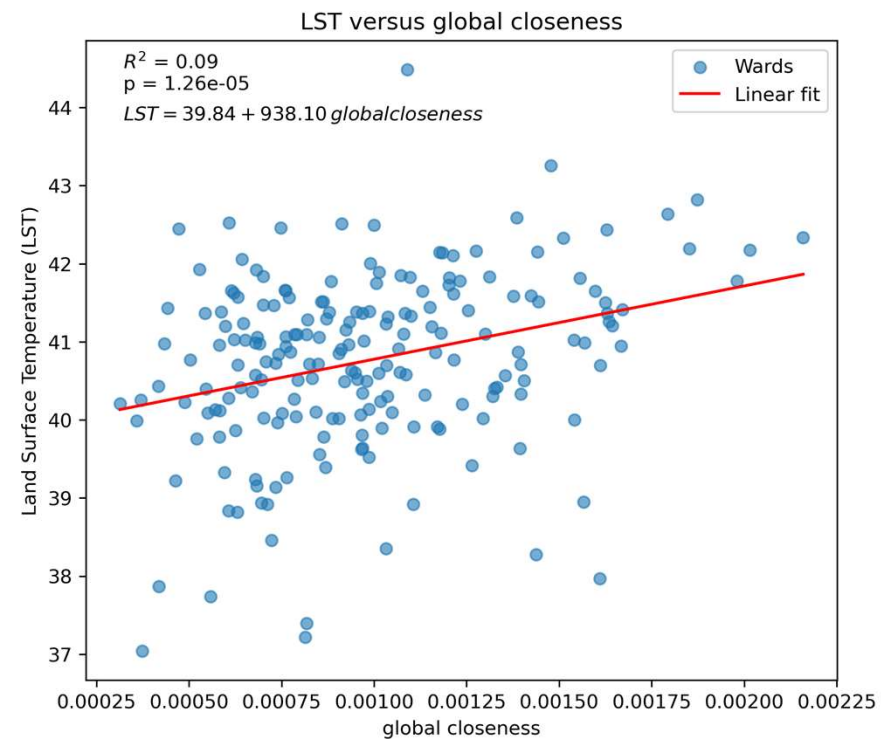


Chennai

Global closeness: Regression

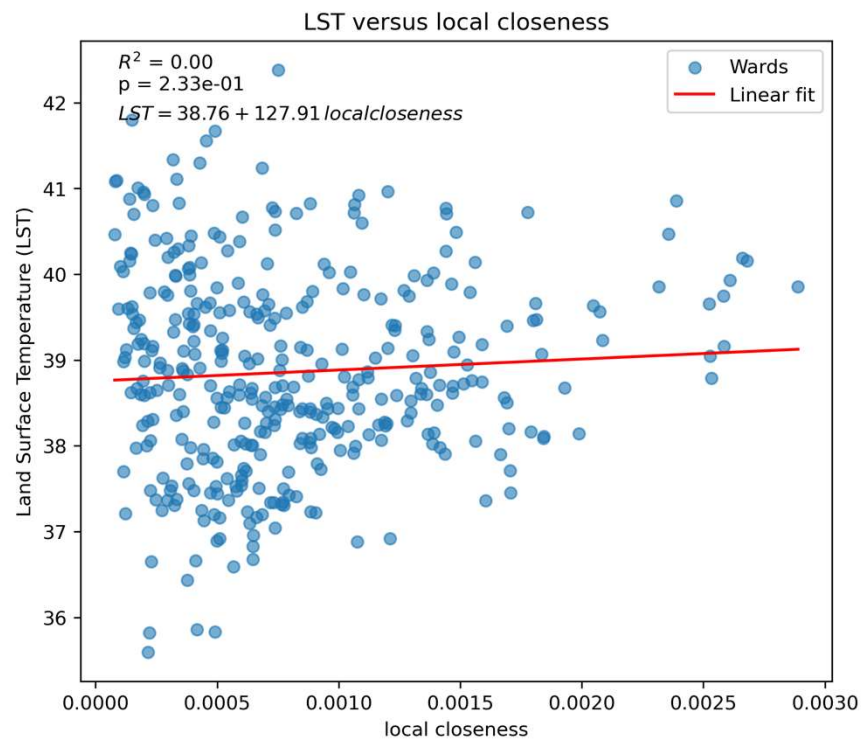


Bangalore

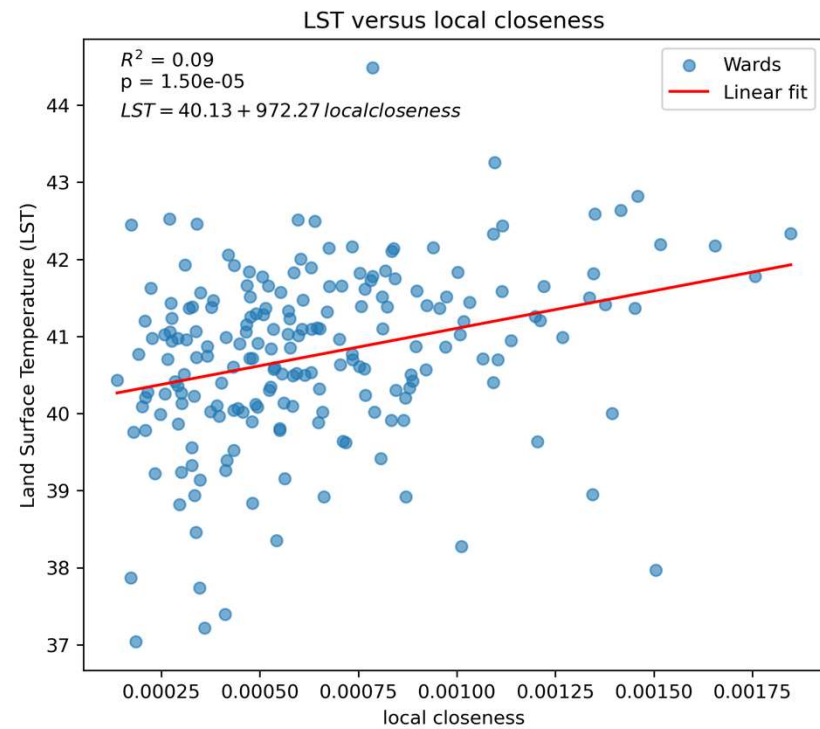


Chennai

Local closeness: Regression



Bangalore



Chennai

Betweenness centrality for edges (e)

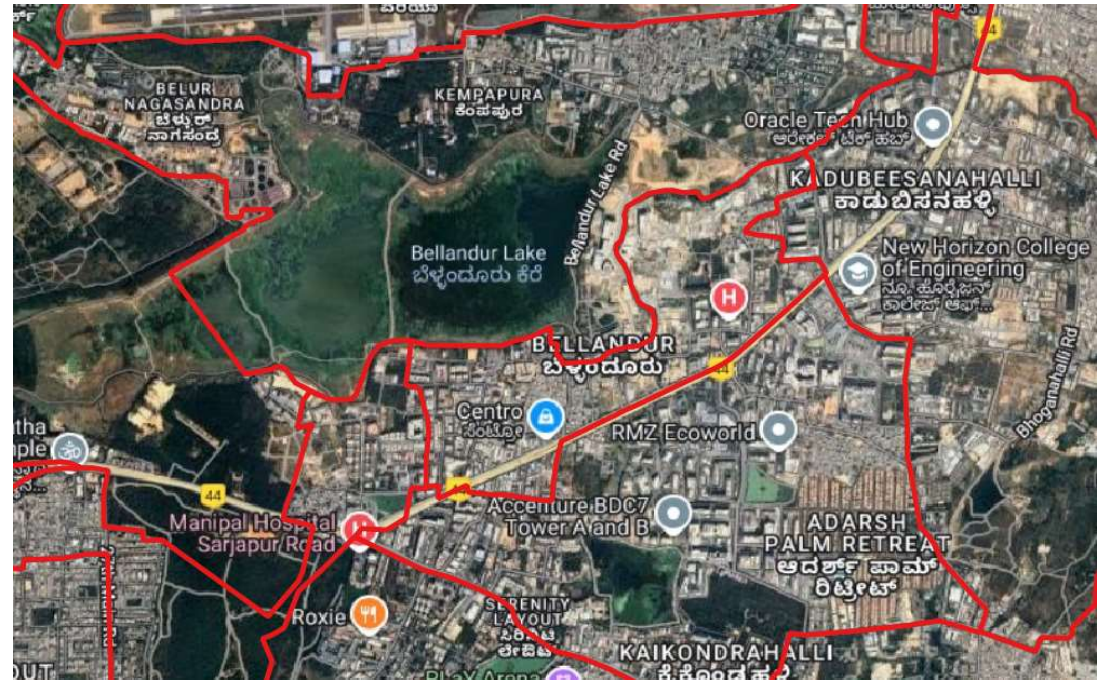
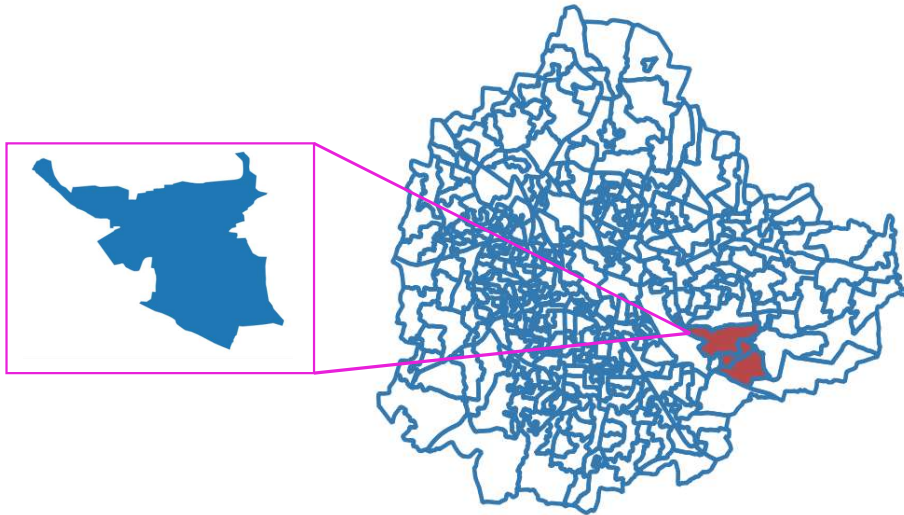
$$c_B(e) = \sum_{s,t \in V} \frac{\sigma(s,t|e)}{\sigma(s,t)}$$

where V is the set of nodes, $\sigma(s,t)$ is the number of shortest -paths, $\sigma(s,t|e)$ and is the number of those paths passing through edge e .

- Measure for the amount of “flow” passing through
- We would expect a positive correlation between betweenness centrality and LST
- Literature has mixed results and some papers obtain a negative relationship (e.g. Guo et al. (2024))
- Can also be computed at the level of the nodes

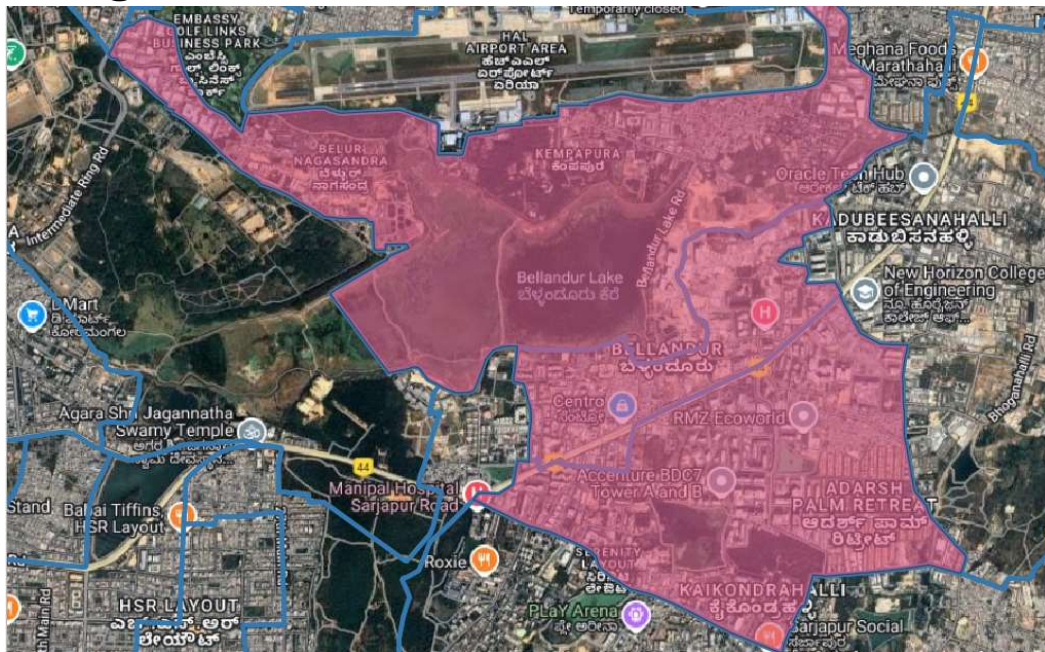
Betweenness centrality – Bellandur area (Bangalore)

- Merge three wards:
 - Yamalur
 - Bellanduru
 - Shivanasamudra

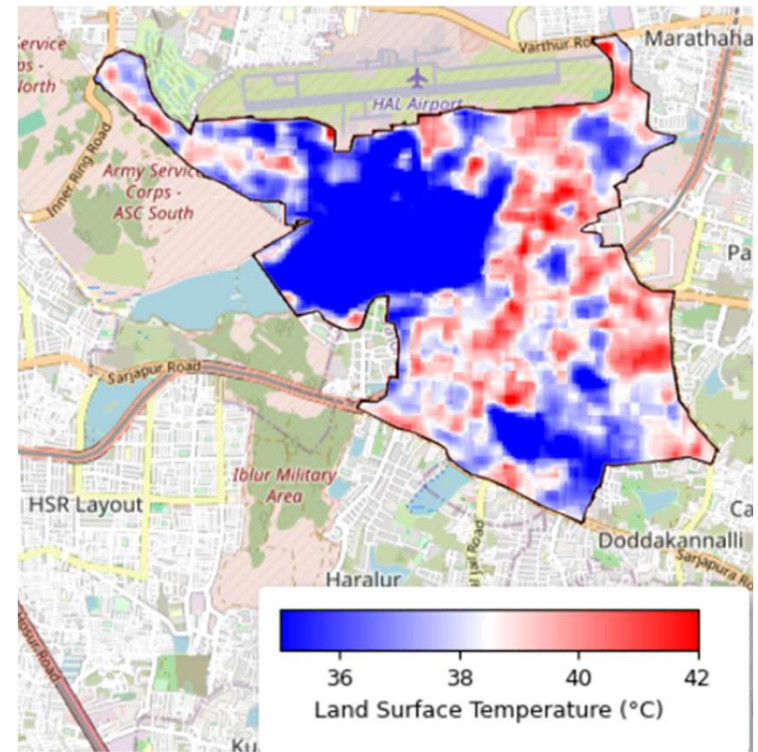
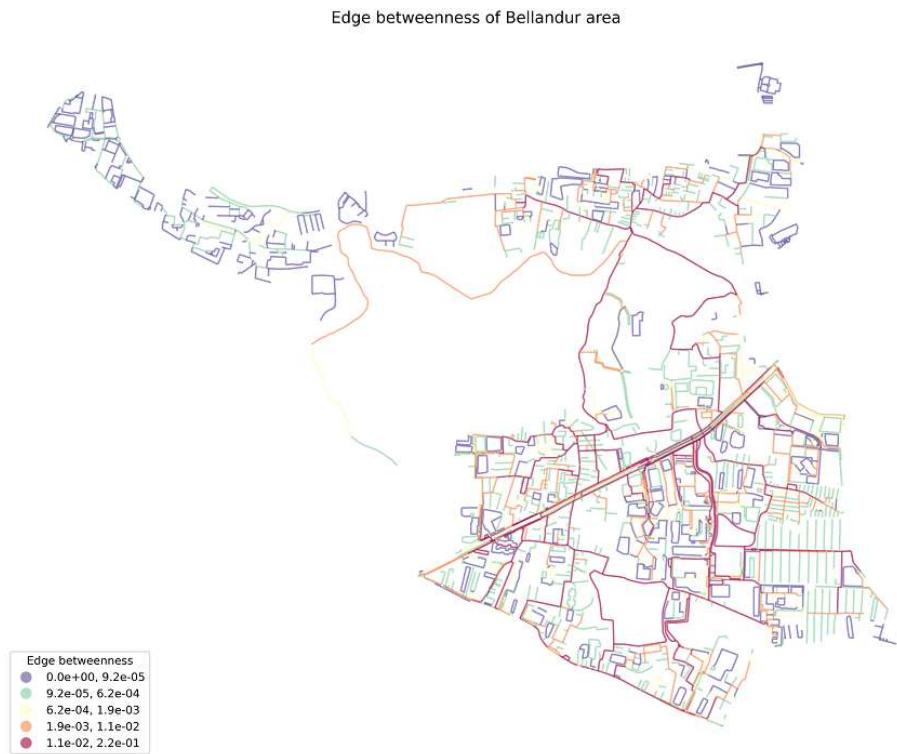


Bellandur area (Bangalore)

- This is the Bellandur area for three wards together
- Lots of IT companies
- Hughe road bottle neck!

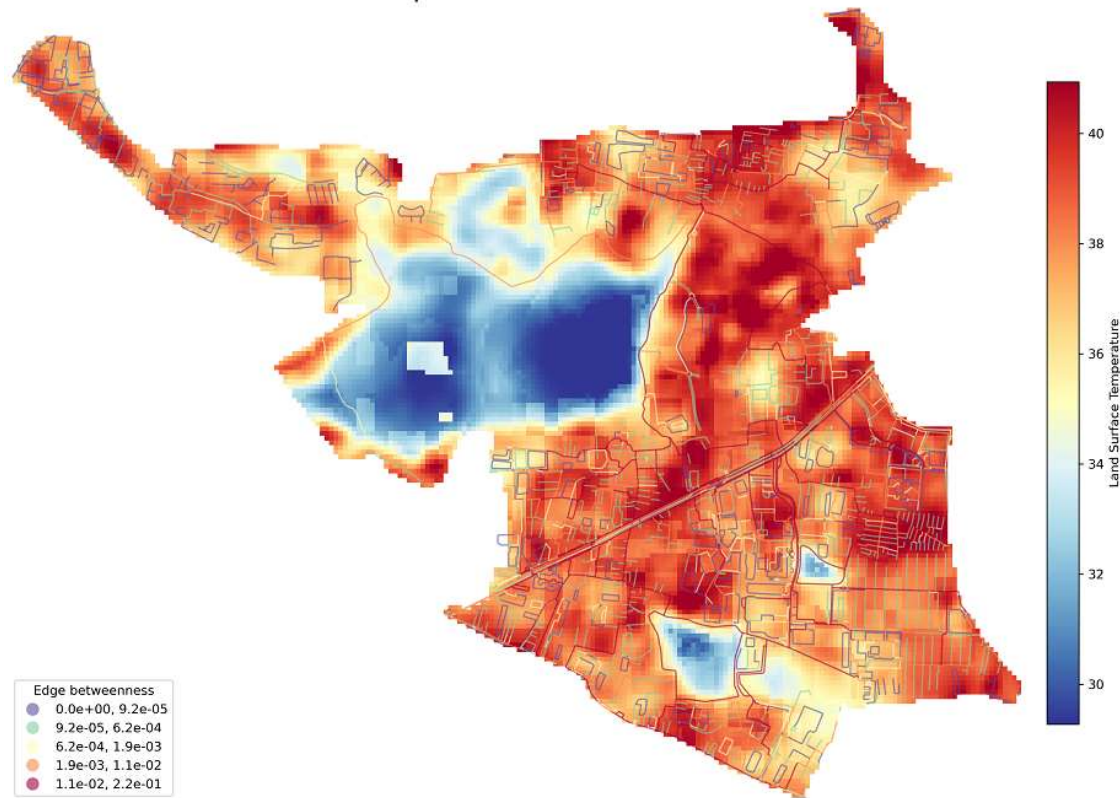


Betweenness centrality and LST



Betweenness centrality

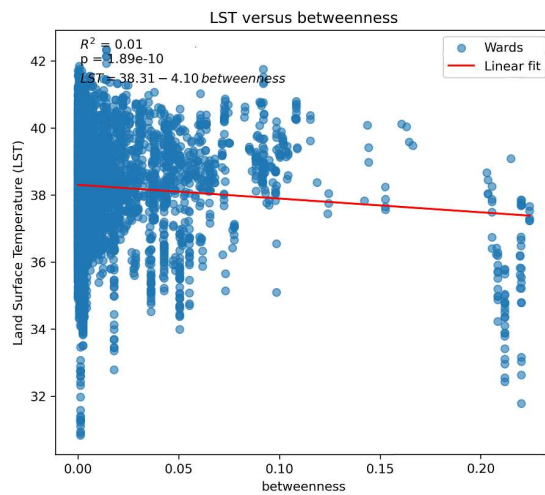
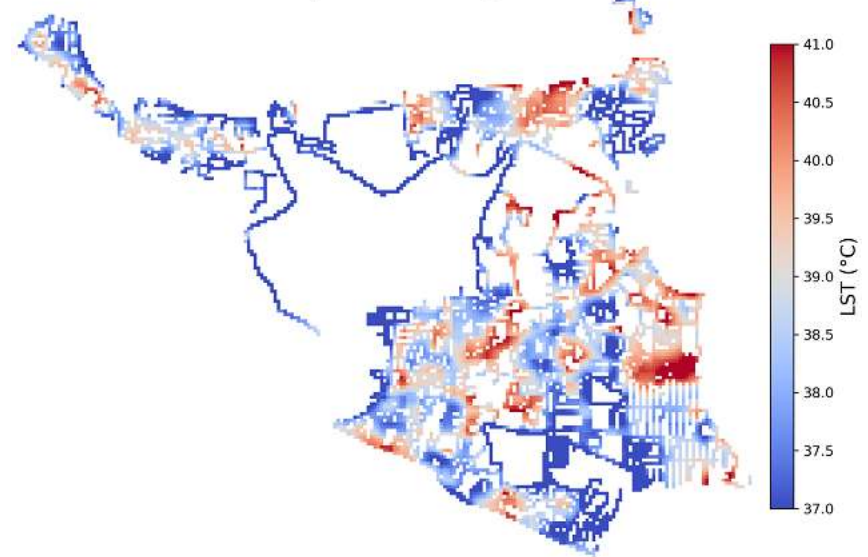
Land Surface Temperature and Road Betweenness



Road betweenness raster



Land Surface Temperature only on road pixels



- There is a slight negative correlation
- The correlation coefficient is -0.08 and it is statistically significant
- Computed on all roads

Conclusion

- NDVI, MNDWI, BSI and NDBI mostly explain the LST
- Some evidence that road network characteristics do matter too
- Results are in general quite robust whether all roads or only drivable roads are used

Next steps

- Analysis at pixel level (30 – 100m) or hexagons
 - Measures like betweenness centrality do not make sense at the ward level
- Night and day – Ahmedabad study (MODIS only for LST but lower resolution)
- Full fledged regression based analysis
 - Incorporation of more variables like height (DEM), slope, building density, etc
- Differences during seasons
- Split up according to the type of road
- (Simplified) Urban Heat Index instead of LST– see Brock (2023) for SUHI
- Node density, intersection density, etc to consider too
- Combinations such as drivable and service roads
- Scaling betweenness centrality to a larger/more areas
- Traffic is not taken into consideration too

Code

- <https://github.com/EllenB/Istroad2>
- This works greatly benefits from the tutorials, software packages, work and scripts of others:
 - [geemap](#) by [Quisheng Wu](#)
 - [Tutorials of Spatial Thoughts](#) by [Ujaval Gandhi](#)
 - [QGIS](#) (occasionally for viewing some maps)
 - [Mapshaper](#) by [Matthew Bloch](#)
 - [osmnx](#) Python package from [Geoff Boeing](#)
 - [Momepy](#) Python package of [Martin Fleis](#)